

KellyLab

How much should you bet? A Kelly-criterion study on simulated data. Yojit Mehrotra. Extends KellyEdge (github.com/YOJITM05/kellyedge). Built with AI assistance.

The formula

With win probability p and net odds b , staking a fixed fraction f of the bankroll gives long-run log-growth $g(f) = p \ln(1 + bf) + (1 - p) \ln(1 - f)$. Setting $g'(f) = 0$ gives $f^* = p - (1 - p)/b$. For $p = 0.55$ and $b = 2$, $f^* = 0.325$, the same as KellyEdge's first test case.

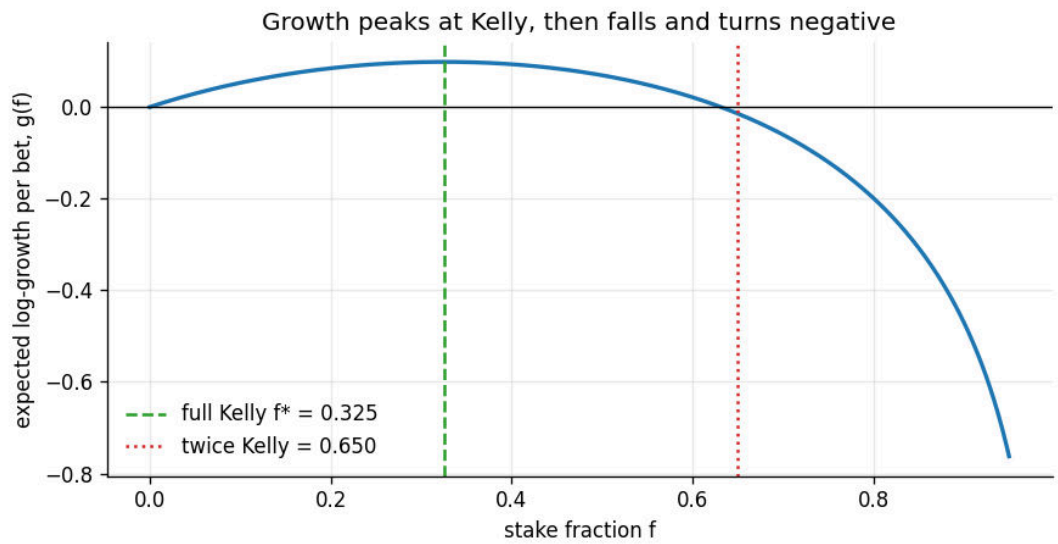


Figure 1. Growth peaks at full Kelly and turns negative at about 1.95 x Kelly.

Simulation: 10,000 bankrolls, 500 bets each

Stake	f	Median growth per bet	Median worst drawdown	Fall below 1% of start	End below start
0.25x Kelly	0.081	0.0447	50%	0.0%	0.0%
0.5x Kelly	0.163	0.0750	78%	0.0%	0.0%
1x Kelly	0.325	0.0986	98%	0.7%	0.0%
1.5x Kelly	0.488	0.0735	100%	20.1%	0.8%
2x Kelly	0.650	-0.0143	100%	87.1%	62.0%

30 simulated bankrolls at three stake sizes

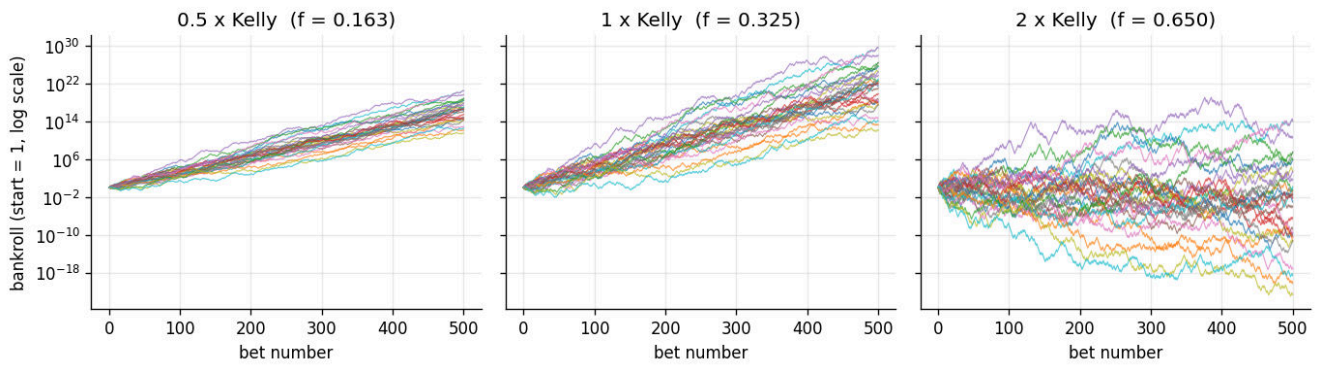


Figure 2. Thirty simulated bankrolls at 0.5x, 1x and 2x Kelly (log scale).

Where the bankrolls end up

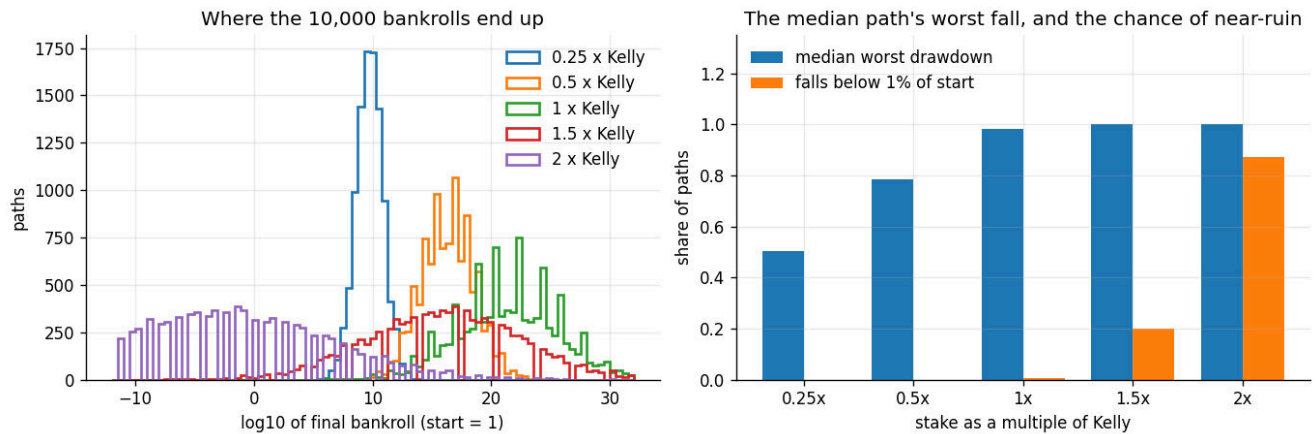


Figure 3. Distribution of final bankrolls, and the median worst drawdown and near-ruin share by stake size.

Estimation error

A trader estimates p from n past trades and bets a fraction k of the estimated Kelly stake. With a strong edge, k near 1 is fine after about 50 observations. With a weak edge ($p = 0.52$, $b = 1$), full estimated Kelly after 100 trades has negative long-run growth and the best k is about 0.3. A rule of thumb is $k = \mu^2 / (\mu^2 + s^2)$, where s is the noise in the estimated edge.

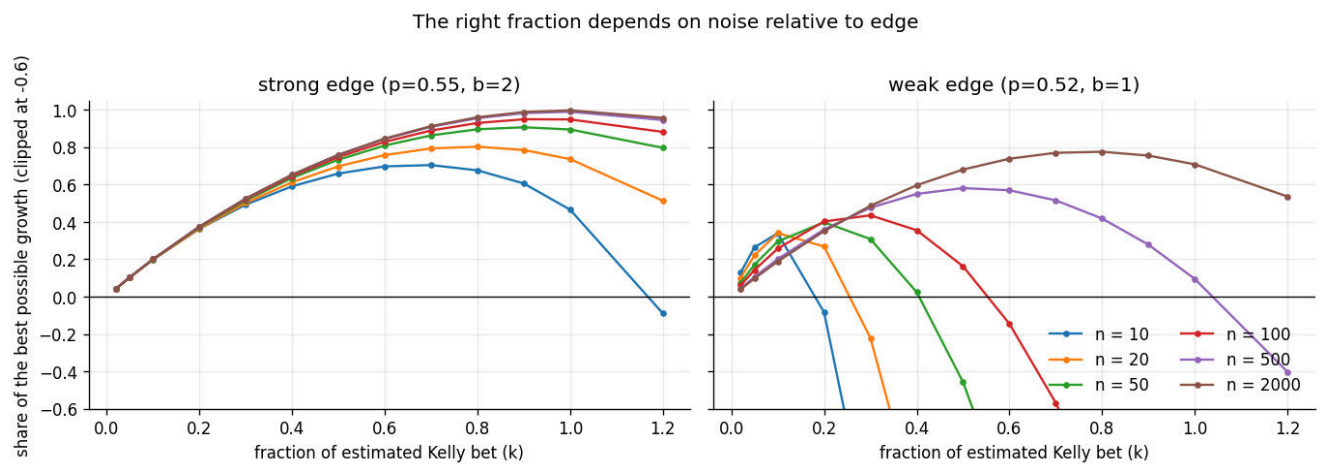


Figure 4. Share of the best possible growth against k , for different numbers of past trades.

Model risk and continuous Kelly

Sized for $p = 0.55$ at full Kelly, growth turns negative if the true p is below 0.440 (half-Kelly: 0.387; quarter-Kelly: 0.360). For 8% mean and 20% volatility a year, the continuous rule μ / σ^2 gives leverage 2.0x.

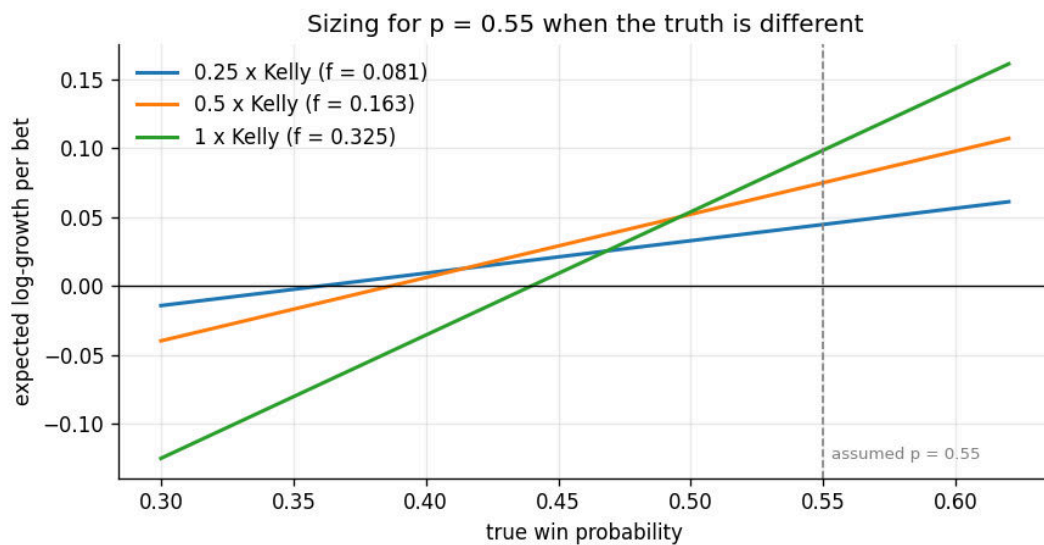


Figure 5. Long-run growth when the true win probability differs from the assumed 0.55.

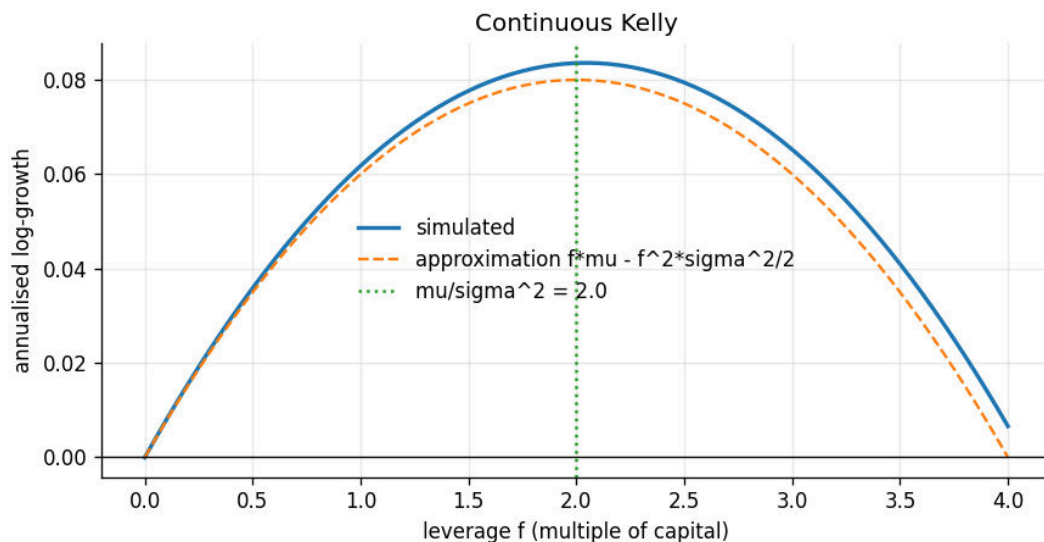


Figure 6. Simulated annualised growth against leverage, with the quadratic approximation.

Limits

- Independent bets, fixed known odds, no costs, one bet at a time.
- Normal returns in the continuous section; real returns have fat tails.
- Simulated data only; not tested on market data; not trading advice.